

## **Nutrient deficiencies in *Brassica napus* modify the ionic composition of plant tissues: a focus on crosstalks between molybdenum and other nutrients**

Anne Maillard, Philippe Etienne, Sylvain Diquélou, Jacques Trouverie, Vincent Billard, Jean-Claude Yvin, and Alain Ourry

### Supplementary data:

Supplemental data SD1. Compositions of the different nutrient solutions derived from Hoagland nutrient solution, used to expose *Brassica napus* L. to individual nutrient deprivations in nitrogen (N), potassium (K), calcium (Ca), phosphorus (P), sulfur (S), magnesium (Mg), iron (Fe), boron (B), manganese (Mn), zinc (Zn), copper (Cu) and molybdenum (Mo).

Supplemental data SD2: Multiple alignment of ASY amino acid sequences from *Arabidopsis thaliana* (At), *Oryza sativa* (Os) and *Brassica napus* (Bna). Orthologues sequences to Os-ASY (BAN228-8) previously characterized (Hibara *et al.* 2013) were retrieved for *Brassica napus* by using blast on the *Brassica napus* genome browser (<http://www.genoscope.cns.fr/blat-server/cgi-bin/colza/webBlat>) and for *Arabidopsis thaliana* by using blast on the non-redundant protein sequences database on the NCBI web site (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). Alignment was performed using Clustal omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>) with At-ASY as reference sequence. Red amino acids and value in front of each sequence correspond to identical amino acids and percentage of identity with the reference sequence, respectively. Bna Sequence name in bold correspond to the one for which gene expression was studied (figure 3).

Supplemental data SD3: Multiple alignment of MOT1 amino acid sequences from *Arabidopsis thaliana* (At), *Oryza sativa* (Os) and *Brassica napus* (Bna). Orthologues sequences to At-MOT1 (gi|15225170) previously characterized (Tomatsu *et al.*, 2007 ; Baxter *et al.*, 2008a) were retrieved for *Brassica napus* by using blast on the *Brassica napus* genome browser (<http://www.genoscope.cns.fr/blat-server/cgi-bin/colza/webBlat>) and for *Oryza sativa* by using blast on the non-redundant protein sequences database on the NCBI web site (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). Alignment was performed using Clustal omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>) with At-MOT1 as reference sequence. Red amino acids and value in front of each sequence correspond to identical amino acids and percentage

of identity with the reference sequence, respectively. Bna Sequence name in bold correspond to the one for which gene expression was studied (figure 3).

Supplemental data SD4: Multiple alignment of CNX1 amino acid sequences from *Arabidopsis thaliana* (At) and *Brassica napus* (Bna). Orthologues sequences to At-CNX1 (gi15242124) previously characterized (Schwarz *et al.* 2000) were retrieved for *Brassica napus* by using blast on the *Brassica napus* genome browser (<http://www.genoscope.cns.fr/blat-server/cgi-bin/colza/webBlat>). Alignment was performed using Clustal omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>) with At-CNX1 as reference sequence. Red amino acids and value in front of each sequence correspond to identical amino acids and percentage of identity with the reference sequence, respectively. Bna Sequence name in bold correspond to the one for which gene expression was studied (figure 3).

Supplemental data SD5: Unrooted phylogenetic tree of amino acid sequences of Mo-Associated proteins (ASY; MOT1 and CNX1) from *Arabidopsis thaliana* (At), *Oryza sativa* (Os) and *Brassica napus* (Bna). Orthologues sequences were retrieved for *Brassica napus* by using blast on the *Brassica napus* genome browser (<http://www.genoscope.cns.fr/blat-server/cgi-bin/colza/webBlat>) and for *Arabidopsis thaliana* and *Oryza sativa* by using blast on the non-redundant protein sequences database on the NCBI web site (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>). Underlined sequences correspond to correspond to the ones for which Bna gene expression was studied (figure 3). Alignment was performed using Clustal omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>) and the phylogenetic tree was build using the Unipro Ugene software (version 1.21.0) with the PHYLIP Neighbour-Joining method.

|                                                                 |         | Concentrations (µM) of nutrients in the solution deprived in: |       |       |       |       |       |       |       |       |       |       |       |
|-----------------------------------------------------------------|---------|---------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Salts                                                           | Control | N                                                             | K     | Ca    | P     | S     | Mg    | Fe    | B     | Mn    | Zn    | Cu    | Mo    |
| CaCO <sub>3</sub>                                               | 2000    | 3250                                                          | 5000  | 0     | 2000  | 2000  | 2000  | 2000  | 2000  | 2000  | 2000  | 2000  | 2000  |
| KNO <sub>3</sub>                                                | 1250    | 0                                                             | 0     | 3750  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  |
| Ca(NO <sub>3</sub> ) <sub>2</sub> , 4H <sub>2</sub> O           | 1250    | 0                                                             | 1875  | 0     | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  |
| KH <sub>2</sub> PO <sub>4</sub>                                 | 250     | 250                                                           | 0     | 250   | 0     | 250   | 250   | 250   | 250   | 250   | 250   | 250   | 250   |
| MgSO <sub>4</sub>                                               | 500     | 500                                                           | 500   | 500   | 500   | 0     | 0     | 500   | 500   | 500   | 500   | 500   | 500   |
| EDTA, 2 NaFe, 3 H <sub>2</sub> O                                | 200     | 200                                                           | 200   | 200   | 200   | 200   | 200   | 0     | 200   | 200   | 200   | 200   | 200   |
| H <sub>3</sub> BO <sub>3</sub>                                  | 10      | 10                                                            | 10    | 10    | 10    | 10    | 10    | 10    | 0     | 10    | 10    | 10    | 10    |
| MnSO <sub>4</sub>                                               | 5       | 5                                                             | 5     | 5     | 5     | 5     | 5     | 5     | 5     | 0     | 5     | 5     | 5     |
| ZnSO <sub>4</sub>                                               | 3       | 3                                                             | 3     | 3     | 3     | 3     | 3     | 3     | 3     | 3     | 0     | 3     | 3     |
| CuSO <sub>4</sub>                                               | 0.7     | 0.7                                                           | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0     | 0.7   |
| (NH <sub>4</sub> ) <sub>6</sub> Mo <sub>7</sub> O <sub>24</sub> | 0.7     | 0.7                                                           | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0.7   | 0     |
| CoCl <sub>2</sub>                                               | 0.1     | 0.1                                                           | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   |
| NiCl <sub>2</sub>                                               | 0.04    | 0.04                                                          | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  | 0.04  |
| SiO <sub>2</sub>                                                | 100     | 100                                                           | 100   | 100   | 100   | 100   | 100   | 100   | 100   | 100   | 100   | 100   | 100   |
| Na <sub>2</sub> O                                               | 38.78   | 38.78                                                         | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 | 38.78 |
| MgCl <sub>2</sub>                                               | 0       | 0                                                             | 0     | 0     | 0     | 300   | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Na <sub>2</sub> SO <sub>4</sub>                                 | 0       | 0                                                             | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 3     | 0.7   | 0     |
| CaCl <sub>2</sub> , 2 H <sub>2</sub> O                          | 1250    | 1250                                                          | 625   | 0     | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  | 1250  |
| H <sub>2</sub> PO <sub>4</sub>                                  | 0       | 0                                                             | 250   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| KCl                                                             | 250     | 250                                                           | 0     | 2000  | 250   | 0     | 250   | 250   | 250   | 250   | 250   | 250   | 250   |
| HCl                                                             | 0       | 0                                                             | 1500  | 0     | 0     | 0     | 0     | 500   | 0     | 0     | 0     | 0     | 0     |
| NaOH                                                            | 0       | 0                                                             | 0     | 0     | 0     | 0     | 0     | 400   | 0     | 0     | 0     | 0     | 0     |
| KOH                                                             | 0       | 1250                                                          | 0     | 50    | 250   | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| KHSO <sub>4</sub>                                               | 0       | 0                                                             | 0     | 0     | 0     | 0     | 500   | 0     | 0     | 0     | 0     | 0     | 0     |

|          |         | Concentrations (µM) of nutrients in the solution deprived in: |        |        |        |        |        |        |        |        |        |        |        |
|----------|---------|---------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Elements | Control | N                                                             | K      | Ca     | P      | S      | Mg     | Fe     | B      | Mn     | Zn     | Cu     | Mo     |
| N        | 3754.2  | 4.2                                                           | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3754.2 | 3750   |
| K        | 1750    | 1750                                                          | 0      | 6050   | 1750   | 1500   | 2250   | 1750   | 1750   | 1750   | 1750   | 1750   | 1750   |
| Ca       | 4500    | 4500                                                          | 7500   | 0      | 4500   | 4500   | 4500   | 4500   | 4500   | 4500   | 4500   | 4500   | 4500   |
| P        | 250     | 250                                                           | 250    | 250    | 0      | 250    | 250    | 250    | 250    | 250    | 250    | 250    | 250    |
| S        | 508.7   | 508.7                                                         | 508.7  | 508.7  | 508.7  | 8.7    | 508.7  | 508.7  | 508.7  | 508.7  | 508.7  | 508.7  | 508.7  |
| Mg       | 500     | 500                                                           | 500    | 500    | 500    | 300    | 0      | 500    | 500    | 500    | 500    | 500    | 500    |
| Fe       | 400     | 400                                                           | 400    | 400    | 400    | 400    | 400    | 0      | 400    | 400    | 400    | 400    | 400    |
| B        | 10      | 10                                                            | 10     | 10     | 10     | 10     | 10     | 10     | 0      | 400    | 400    | 400    | 400    |
| Mn       | 5       | 5                                                             | 5      | 5      | 5      | 5      | 5      | 5      | 5      | 0      | 5      | 5      | 5      |
| Zn       | 3       | 3                                                             | 3      | 3      | 3      | 3      | 3      | 3      | 3      | 3      | 0      | 3      | 3      |
| Cu       | 0.7     | 0.7                                                           | 0.7    | 0.7    | 0.7    | 0.7    | 0.7    | 0.7    | 0.7    | 0.7    | 0.7    | 0      | 0.7    |
| Mo       | 4.9     | 4.9                                                           | 4.9    | 4.9    | 4.9    | 4.9    | 4.9    | 4.9    | 4.9    | 4.9    | 4.9    | 4.9    | 0      |
| Ni       | 0.04    | 0.04                                                          | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   | 0.04   |
| Cl       | 2750.28 | 2750.3                                                        | 2750.3 | 2000.3 | 2750.3 | 2800.3 | 2750.3 | 3250.3 | 2750.3 | 2750.3 | 2750.3 | 2750.3 | 2750.3 |
| Co       | 0.1     | 0.1                                                           | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    |
| Si       | 100     | 100                                                           | 100    | 100    | 100    | 100    | 100    | 100    | 100    | 100    | 100    | 100    | 100    |
| Na       | 477.6   | 477.6                                                         | 477.6  | 477.6  | 477.6  | 477.6  | 477.6  | 477.6  | 477.6  | 487.6  | 483.6  | 479.0  | 477.6  |

|   |                    |        |                                                                                      |     |
|---|--------------------|--------|--------------------------------------------------------------------------------------|-----|
| 1 | At-ASY-gi 15217708 | 100.0% | MEIFYFVVFVGGGLAAIVAGLELSKSNKDRINTSSAFNSFKNNYLLVYSLMMAGDWLQGPVYVYLYSTYGFGKGIGQLFIA    | 80  |
| 2 | BnaC01g20370D      | 83.0%  | MEVFYVVFVGVVGLIVVAILELSKNNKDRINTSSAFNSFKNNYLLVYSLMMAGDWLQGPVYVYLYSTYGYKGDIGQLFIA     |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | MEVFYVVFVGGGLAAIVAGLELSKSGKDRVATTEAFNAFKNNYLLVYSLMMSGDWLQGPVYVYLYSYGFGDKGDIGQLFIA    |     |
| 4 | BnaC03g52510D      | 74.5%  | -----MVFCVMAAVVAGLELSKTNKDRINTSSAFNSFKNNYLLVYSLMMAGDWLQGPVYVYLYSTYGFGKGIGQLFIA       |     |
| 5 | BnaA06g16000D      | 71.5%  | MEVFYVVFVGVVMAAVVALEMSKRSKDNQNTSS--FNSFKNNYLLVYSLMMAGDWLQGPVYVYLYSTYGFGKRDIGQLFIA    |     |
| 6 | BnaC08g21010D      | 29.7%  | MKIDPPPIKSVLYCAYYGSRITLSGIEREAKVTEMEVFYYLVFGVMAAVVAALEMSKRKNDQNTSSPFNSFKNSYVLFVFSVMM |     |
| 1 | At-ASY-gi 15217708 | 100.0% | GFGSSMLFGTIVGSLADKQGRKRASITYCITYILSCITKHSPQYKVLVGRVLGGIATSLIFSASFESWLVAEHNKRGFEQ     | 160 |
| 2 | BnaC01g20370D      | 83.0%  | GFGSSMLFGTIVGSLADKQGRKRACITYCITYILSCITKHSPQYKVLVGRVLGGIATSLIFSASFESWLVAEHNKRGFEQ     |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | GFGSSMLFGTIVGSLADKQGRKRACITYCITYILSCITKHSPYKRLMLVGRVLGGIATSLIFSASFESWLVAEHNKRGFD     |     |
| 4 | BnaC03g52510D      | 74.5%  | GFGSSMLFGTIVGSLADKQGRKRACITYCITYILSCITKHSPQYKVLVGRVLGGIATSLIFSASFESWLVAEHNKRNFD      |     |
| 5 | BnaA06g16000D      | 71.5%  | GFGSSMLFGTIVGSLADKQGRKRACITYCITYILSCITKHSPQYKVLVGRVLGGIATSLIFSASFESWLVAEHNKRNFD      |     |
| 6 | BnaC08g21010D      | 29.7%  | AGDWLQGPVYVYLYSTYGGKRDIGQLFIAGFGSSMLFGTIVGSLAHKQGLILSVCVLDSGVERGHVYILYSLHSCITK       |     |
| 1 | At-ASY-gi 15217708 | 100.0% | QWLSVTFSKAIFLGNGLVAIIAGLFGNYLVDSLGLPVAPFDAAACFLAIGMAVISSWSSENYGDPSENKDLLTQFKNAA      | 240 |
| 2 | BnaC01g20370D      | 83.0%  | QWLSVTFSKAVFEGNGLVAIIAGLFGNLLVHSSSLGPVAPFDAAACFLAIGMAVISSWSSENYGDPSENKDLLTQFERGAA    |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | QWLSVTFSKAIFLGNGLVAIIAGLFGNLLADNIGFVPVAPFDAAACFLAIGMAVISSWSSENYGDPSENKDLLAQFKVAA     |     |
| 4 | BnaC03g52510D      | 74.5%  | QWLSVTFSKAVFLGNGLVAIIAGLFGNLLVDTFSGFVPVAPFDAAACFLAIGMAVILTTWSSENYGDPSENKDLLTQFKVAA   |     |
| 5 | BnaA06g16000D      | 71.5%  | QWLSVTFSKAVFEGNGLVAIIAGLFGNLLVDTFSGFVPVAPFDAAACFLAIGMAVILTTWSSENYGDPSENKDLLTQFKVAA   |     |
| 6 | BnaC08g21010D      | 29.7%  | HSPQNTVLMVGRILGLIATSLNLSAFESWLVAEHNKLLSLTISKTVFEGNGLVAIIAGLFGNLLVDVFSFGPVAPFDAAAC    |     |
| 1 | At-ASY-gi 15217708 | 100.0% | SAIAS-----DEKIALLGAIQSLFEGSMYTFVFLWTPALS-----PNEEDIPHGFI FATFML                      | 320 |
| 2 | BnaC01g20370D      | 83.0%  | VAIAS-----DEKIALLGAIQSLFEGSMYTFVFLWTPALS-----PNDEEIPHGFI FATFML                      |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | KAIAS-----DEKIALLGAIQSLFEGSMYTFVFLWTPALS-----PNDEEIPHGFI FATFML                      |     |
| 4 | BnaC03g52510D      | 74.5%  | IAIASGSKTVSVYSPLSLTQSMDDFDEKIALLGAIQSLFEASMYTFVFLWTPALS-----PNDEEIPHGFI FATFML       |     |
| 5 | BnaA06g16000D      | 71.5%  | IAIASGRFKVN-----VKNYDEKIALLGAIQSLFEASMYTFVFLWTPALS-----PNDEEIPHGFI FATFML            |     |
| 6 | BnaC08g21010D      | 29.7%  | CFIAI-----GMVITLTTWSSENYGDPSENKDLLMTQFKVAAIAIASGSPNEEIPHGFI FATFML                   |     |
| 1 | At-ASY-gi 15217708 | 100.0% | ASMLGSSIASRLLAHSSPKVES-----YMQIVFVISSAALMLPVVTSFLVAPSGVKGGSISFSGCIQLMGFCTFEACV       | 400 |
| 2 | BnaC01g20370D      | 83.0%  | ASMLGSSIASRLLSRSSPKVES-----YMQIVFLVSAASLLLPITMASLVAPSVKGGGIFSGGCIQLMGFCTFEACV        |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | SSMLGSSIASRLLARKL-KVEG-----YMQIVFVISAFTLFLPVVTHFLVFPSSVKGGGIFSGGCIQLMGFCTFEACV       |     |
| 4 | BnaC03g52510D      | 74.5%  | ASMLGSSLAARLMARSSLRVEN-----YMQIVFLVSAASLLLPITTSVLVTPSRVKEEGISLTCSIQLLGFCVFEACI       |     |
| 5 | BnaA06g16000D      | 71.5%  | ASMLGSSLAARLMARSSLRVEK-----YMQIVFLVSAASLLLPITTSVLVTPSRVKEEGISLTCSIQLLGFCVFEACI       |     |
| 6 | BnaC08g21010D      | 29.7%  | ASMLGSSLAARLMARSSLRVEN-----YMQIVFLVSAASLLLPITTSVLVTPSRVKEEGISLTSSIQLLGFCVFEACI       |     |
| 1 | At-ASY-gi 15217708 | 100.0% | GIFWPSIMKMRSQYIPEEARSTIMNFFRPLNIFVCLVLYNVDAFPMTVMFGMCSVFLFVASILQRRLLMNVAE--IHKSRS    | 480 |
| 2 | BnaC01g20370D      | 83.0%  | GIFWPSIMKMRSQYIPEEARSTIMNFFRPLNIFVCLVLYNVNEFPMTVMFGMCSVFLFVASILQRRLLMNVAD--KEKAND    |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | GIFWPSIMKMRSQYIPEEARSTIMNFFRPLNIFVGVLYNVNAFEITVMFGMCATFLFMAAILQRRLLMNVSDLHRSTKA      |     |
| 4 | BnaC03g52510D      | 74.5%  | GIFWPSIMKMRSQYIPEEARSTIMNFFRPLNIFVCLVLYNVDAFEITVMFGMCSVFLFVASILQRRLLMNVIS-----       |     |
| 5 | BnaA06g16000D      | 71.5%  | GIFWPSIMKMRSQYIPEEARSTIMNFFRPLNIFVCLVLYNVDAFEITVMFGMCSVFLFVASILQRRLLMNVIS-----       |     |
| 6 | BnaC08g21010D      | 29.7%  | GIFWPSIMKMRSQYIPEEARSTIMNFFRPLNIFVCLVLYNVDAFEITVMFGMCSVFLFVASILQRRLLMNVIS-----       |     |
| 1 | At-ASY-gi 15217708 | 100.0% | QEWSAEKEMTSEADPLNP-----                                                              | 511 |
| 2 | BnaC01g20370D      | 83.0%  | WTPLNERNAAEAPLNI-----                                                                |     |
| 3 | Os-ASY-BAN20028.1  | 80.7%  | VEMTGDEPLNP-----                                                                     |     |
| 4 | BnaC03g52510D      | 74.5%  | -EKPKGEEWSPMKDRNSEVDPLTL-----                                                        |     |
| 5 | BnaA06g16000D      | 71.5%  | SENPKGEEWSPMKEDNSEVDALIL-----                                                        |     |
| 6 | BnaC08g21010D      | 29.7%  | ENPKGEEPLSPMKERNSEVDPLIMFLSQSK-                                                      |     |



|                                     |                      |        |                                                                                   |
|-------------------------------------|----------------------|--------|-----------------------------------------------------------------------------------|
| 1                                   | At-CNX1-gi 15242124  | 100.0% | MEGQGCCGGGGGKTEMIPTTEALRIVFGVSKRLPPVIVSLYEALGKVLAEDIRAPDPLPPYPASVKDGYAVVASDGPGEY  |
| 2                                   | BnaA03g08200D        | 89.0%  | MDASEGCGSGDEKSEMIPTTEALKIVLSVSERLAPVIVSLQEALGKVLAEDIRAPDPLPPYPASVKDGYAVVASDGPGEY  |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | MDASEGCGSGDEKSEMIPTTEALRIVLSVSERLPPVIVSLQEALGKVLAEDIRAPDPLPPYPASVKDGYAVVASDGPGEY  |
| 81 . 1 . . . . :                    |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | PVITESRAGNDGLGVTVP TPGTVAYVTGGPIPDGADAVVQVEDTKVIGDVSTESKRVKILIQTKKGTDIRRVGCDIEKDA |
| 2                                   | BnaA03g08200D        | 89.0%  | PVITESRAGNDGLGVTVP TPGTVAYVTGGPIPDGADAVVQVEDTKVIGDVSTESKRVKILIQTKKGTDIRRVGCDIEKDA |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | PVITESRAGNDGLGVTVP TPGTVAYVTGGPIPDGADAVVQVEDTKVIGDVSTESKRVKILIQTKKGTDIRRVGCDIEKDA |
| 161 . . . . 2 . . . . 240           |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | TVLTTGERIGASEIGLLATAGVTLMVKVYPMPIVAILSTGDELVEPTAGTLGRGQIRDSNRAMLVAAVMQQCKVVDLGIV  |
| 2                                   | BnaA03g08200D        | 89.0%  | TVLKTGERIGASEIGLLATAGVTLMVKVYPMPIVAILSTGDELVEPTAGTLGRGQIRDSNRAMLVAAVMQQCKVVDLGIV  |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | TVLKTGERIGASEIGLLATAGVTLMVKVYPMPIVAILSTGDELVEPTAGTLGRGQIRDSNRAMLVAAVMQQCKVVDLGIV  |
| 241 : . . . . 3 . . . . 320         |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | RDDRKELEKVLDEAVSSGVDIILTSGGVSMGDRDFVKPLLEEKGVYFSKVLMPGKPLTFAETIRAKPTESMLGKTVLAF   |
| 2                                   | BnaA03g08200D        | 89.0%  | RDDRKELEKVLDEAVSSGVDIILTSGGVSMGDRDFVKPLLEEKGVYFSKVLMPGKPLTFAETIRAKPTESMLGKTVLAF   |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | RDDRKELEKVLDEAVSSGVDIILTSGGVSMGDRDFVKPLLEEKGVYFSKVLMPGKPLTFAETIRAKPTESMLGKTVLAF   |
| 321 . . . . : . . . . 4 400         |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | GLPGNPVSLVCFNIFVVP TIRQLAGWTS PHPLRVLRRLQEPKSDPIRPEFHRAIIKWKDNDGSGTSGFVAESTGHQMS  |
| 2                                   | BnaA03g08200D        | 89.0%  | GLPGNPVSLVCFNIFVVP TIRQLAGWTS PHPLRVLRRLQEPKSDPIRPEFHRAIIKWKDNDGSGTSGFVAESTGHQMS  |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | GLPGNPVSLVCFNIFVVP TIRQLAGWTS PHPLRVLRRLQEPKSDPIRPEFHRAIIKWKDNDGSGTSGFVAESTGHQMS  |
| 401 . . . . : . . . . 480           |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | SRLLSMRSANALLELPATGNVLSAGSSVSAIIVSDISAFSIDKKASLSEPGSIRKEKKYDEVPGPEYKVAILTVDTSVA   |
| 2                                   | BnaA03g08200D        | 89.0%  | SRLLSMRSANALLELPATGNVLSAGTSVSAIIVSDISAFSIDKKASLSEPGSIRKEKKYDEVPGPEYKVAILTVDTSVA   |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | SRLLSMRSANALLELPATGNVLSAGTSVSAIIVSDISAFSIDKKASLSEPGSIRKEKKYDEVPGPEYKVAILTVDTSVA   |
| 481 . . . . 5 . . . . : . . . . 560 |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | GAGPDRSGPRAVSVVDSSEKLGGAQVAVAVVPDEVERIKDILQKWSVDVEMDLILTLLGGTGFTPRDVTPEATKKVIER   |
| 2                                   | BnaA03g08200D        | 89.0%  | GAGPDRSGPRAVSVVDSSEKLGGAQVAVAVVPDEVERIKDILQKWSVDVEMDLILTLLGGTGFTPRDVTPEATKKVIER   |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | GAGPDRSGPRAVSVVDSSEKLGGAQVAVAVVPDEVERIKDILQKWSVDVEMDLILTLLGGTGFTPRDVTPEATKKVIER   |
| 561 . . . . 6 . . . . . 640         |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | ETPGLLFVMMQESLKITPFAMLSRSAAGIRGSTLIINMPGNPNAVAECMEALLPALKHALKQIKGDKREKHPKHIPHAEA  |
| 2                                   | BnaA03g08200D        | 89.0%  | ETPGLLFVMMQESLKITPFAMLSRSAAGIRGSTLIINMPGNPNAVAECMEALLPALKHALKQIKGDKREKHPKHIPHAEA  |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | ETPGLLFVMMQESLKITPFAMLSRSAAGIRGSTLIINMPGNPNAVAECMEALLPALKHALKQIKGDKREKHPKHIPHAEA  |
| 641 : . . . . 1 670                 |                      |        |                                                                                   |
| 1                                   | At-CNX1-gi 15242124  | 100.0% | TLPTDTWDQSYKSAYETGEKKEAGCSCTH                                                     |
| 2                                   | BnaA03g08200D        | 89.0%  | TLPTDTWDQSYKSAYETGEKKEAGCSCTH                                                     |
| 3                                   | <b>BnaC03g10400D</b> | 86.9%  | SPAPAADTWDKSYKAAAYEGAEETKDAGCSC                                                   |

